A Blockchain-Based Framework for Ethereum Phishing Detection Using Machine Learning, Deep Learning, and Explainable Artificial Intelligence

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ABSTRACT

Phishing attacks have emerged as a major security concern in blockchain technology due to its decentralized nature and immutable cryptocurrency transaction system. Ethereum is one of the most widely used platforms for executing smart contracts and developing decentralized applications. This platform is also vulnerable to phishing attacks due to wallet address manipulation, contract interaction, and social engineering attacks. The loss due to these malicious transaction attacks is irreversible once executed in the blockchain system, making it crucial to develop efficient detection methods for analyzing the transaction data. This paper proposes a state-of-the-art Ethereum phishing detection system using classical machine learning (ML), deep learning (DL), and explainable artificial intelligence (XAI). Features such as transaction value, block height, and timestamp are derived from real-world Ethereum transaction datasets. Temporal features are also considered, including hour, day, and month, which are relevant to detecting the behavioral patterns of phishing attacks. Several machine learning algorithms, including Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine, K-Nearest Neighbors, and a feedforward network to learn complex non-linear relationships in transaction patterns, are implemented and compared. In order to ensure practical relevance, experiments are performed on balanced datasets for controlled analysis and imbalanced datasets, which are closer to real-world transaction distributions. In addition, performance is evaluated using precision, recall, F1-score, and ROC-AUC. From the experimental analysis, it is evident that ensemble learning techniques and the deep learning model provide robust detection capabilities, especially in imbalanced conditions. In order to mitigate the interpretability limitations of predictive models, SHAP (SHapley Additive exPlanations) is used to provide global and local interpretability of model predictions. From the explainability analysis, it is evident that transaction value and temporal characteristics are strong phishing indicators. By combining predictive accuracy with interpretability, the proposed framework offers a practical and trustworthy solution for blockchain security monitoring and intelligent fraud prevention in decentralized financial systems.

Keywords: Blockchain, Ethereum, Phishing Attack, Machine learning, Deep learning, SHAP

1. Introduction

The proposed framework, which leverages both predictive accuracy and interpretability, can thus provide a useful solution for blockchain security monitoring and smart fraud prevention in decentralized financial systems. Blockchain technology has revolutionized digital transactions by enabling decentralized, transparent, and tamper-resistant record-keeping without the need for any centralized intermediary [2, 3]. Platforms such as Ethereum have further enhanced the functionality of blockchain by enabling smart contracts, decentralized applications, and decentralized finance, which can be leveraged for various use cases such as decentralized finance, digital exchange, and tokenized systems [4, 7]. Despite the various advantages offered by blockchain, the decentralized and pseudonymous nature of blockchain networks have also introduced several security challenges to the community [3, 5]. Among them, phishing attacks have become one of the most common and financially damaging threats to blockchain users [4, 8, 9].

The phishing in the context of the blockchain is different from the phishing in the context of the traditional online environment because of the permanent and immutable nature of the cryptocurrency transactions [2]. The attackers take advantage of the users by providing them with fake wallet addresses, malicious smart contracts, and fake websites. The attackers also employ various social engineering techniques that enable them to trick the users into sending unauthorized transactions [4], [9]. Once the transactions are confirmed in the blockchain environment, they are permanent and immutable. Therefore, the users suffer permanent loss [3]. The Ethereum blockchain environment is highly popular and has the highest number of transactions and smart contracts. Therefore, the attackers are highly interested in performing scams in the context of the Ethereum blockchain [8], [9]. The attackers' dynamic and changing behavior makes [3, 10].

To mitigate the challenges, data-driven methods using machine learning (ML) and deep learning (DL) have emerged as potential solutions to detect phishing attacks [1, 5, 6]. These models can process large-scale transactional data to automatically learn patterns that can differentiate between legitimate and illegitimate transactions [1, 8]. Various features of the transactions, such as the value of the transactions, blocks, and temporal information, can provide valuable insight into the phishing attacks by showing their signatures [5, 10]. Classical machine learning models, such as decision trees, ensemble models, and support vector machines, can provide efficient classification for detecting phishing attacks [6, 10]. Moreover, deep learning models can learn complex non-linear relationships between features

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to improve the detection of phishing attacks in dynamic environments [1, 9].

However, despite the accuracy and ability to make predictions, ML and DL models are often criticized for lacking interpretability. This is particularly important in security-related fields like blockchain fraud detection, as it is crucial for various stakeholders, including developers and regulatory bodies, to understand and be able to explain the decision-making process in automated systems. This has led to the integration of XAI methods, which provide information about model behavior and highlight key features that contribute to predictions. One XAI method is SHAP (Shapley Additive exPlanations), which is grounded in theory and allows for feature importance to be quantified at both global and local levels. [5, 3].

With this in mind, the present work proposes an interpretable Ethereum phishing detection system that utilizes classical ML models, deep learning, and SHAP explainability. The system utilizes transactional and time-based feature sets in phishing detection while ensuring the transparency and interpretability of the phishing detection process. The proposed system is expected to improve the security of blockchain systems and deploy intelligent phishing detection systems in an interpretable manner [1, 3].

The main contributions of this study are as follows:

- A blockchain-based phishing detection framework that utilizes ML, DL, and SHAP-based XAI techniques.
- A comparative analysis of the performance of five traditional ML models and a feedforward neural network for both balanced and imbalanced dataset conditions.
- The utilization of SHAP-based explainability for global feature importance analysis and local explanation of predictions.
- The analysis of temporal and transactional features as indicators of phishing behavior on the Ethereum blockchain.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the problem formulation. Section 4 describes the proposed framework. Sections 5–7 discuss the dataset, preprocessing, and feature engineering. Sections 8 and **??** present the machine learning and deep learning models. Section 10 describes the experimental setup, followed by evaluation metrics in Section 11. Section 12 presents results and performance analysis. Section 13 discusses explainability using SHAP. Sections 14–16 present comparative discussion, limitations, and future work. Finally, Section 17 concludes the paper.

2. Related Work

Blockchain fraud and phishing detection have gained significant research interest as the usage of cryptocurrencies grows and financial risks escalate [2, 3]. Initially, detection mechanisms focused on rule-based and heuristic detection, where simple transactional features, such as abnormal value transactions, transaction frequency, and address blacklisting, are considered [10]. Although these mechanisms offered initial defense, they are not sufficient to cope with the evolving nature of phishing attacks and the scalability of blockchain networks, where attackers are constantly updating their behaviors [3, 5].

To mitigate such issues, supervised machine learning models have been used for the analysis of blockchain transactions, as proposed in several recent research works [5, 6, 10]. Classical machine learning models such as Decision Trees, Random Forest, Support Vector Machines, and Gradient Boosting have been used to detect anomalous patterns in the blockchain environment using features such as transaction value, block information, temporal activity, and address interaction [5, 10]. Ensemble machine learning models have also shown promising results for fraud detection problems due to their robustness in dealing with non-linear relationships in the data [1, 5]. However, such models are often dependent on manual feature engineering and may not be effective in dealing with complex temporal relationships in phishing attacks [3].

Studies have been conducted to investigate the potential of applying deep learning models in cryptocurrency fraud detection [1, 2, 9]. Neural networks, including feedforward networks and their variants, were shown to perform well in cryptocurrency fraud detection by automatically learning hierarchical feature representations from the vast amount of transaction data [1, 8]. This is particularly important in detecting potential fraud, where the ability of the ML models to learn the complex patterns of the cryptocurrency users is critical, especially in cases where the attacks are constantly changing and involve coordinated attacks using phishing [9]. However, the main disadvantage of using deep learning models is their black box nature, which is considered a critical disadvantage in building trust among the stakeholders involved [2, 3].

In order to mitigate the interpretability issue, techniques of Explainable Artificial Intelligence (XAI) are integrated into the system [3, 5]. XAI techniques include SHAP and LIME, which provide explanations through the estimation of the contribution of each feature to the prediction [5]. SHAP is unique in providing a theoretical basis using the concept of Shapley values, which enables both global and local explanations [5]. XAI techniques are applied in blockchain security to provide insights into the impact of transactional attributes, which enables the understanding of suspicious activities and validation of the decision-making process [3, 5].

Another class of methods that has received significant traction is graph-based methods. Zhang et al. [4] presented a graph convolutional network to detect phishing nodes in the Ethereum transaction graph based on the structural properties of address interaction graphs. Kim et al. [8] extended the aforementioned method in the context of phishing account detection in the Ethereum blockchain system using a heterogeneous transaction graph and graph learning methods.

Table 1
Comparison of Blockchain Fraud Detection Studies

Study	Year	Data Type	Method	Key Contribution
Wang et al. [1]	2025	Graph	Dynamic GNN	Detects illegal transactions using temporal graph learning
Zhang et al. [4]	2023	Ethereum graph	GCN	Detects phishing nodes using graph structure
Kim et al. [8]	2023	Graph	Heterogeneous GNN	Improvised phishing detection
Chen et al. [9]	2024	Ethereum	Hybrid GNN	Handles data imbalance and phishing detection
Pocher et al. [5]	2023	Bitcoin	ML + Graph	AML fraud detection framework
Asiri et al. [6]	2025	Bitcoin	GCN	Fraud detection using graph learning
Kasula et al. [7]	2025	Ethereum	GNN	Address classification and security analytics
Kılıç et al. [10]	2021	Ethereum	ML	Blacklisted address detection
Shevchuk et al. [3]	2025	Survey	Review	Research trends and challenges
Proposed Work	2026	Transactions	ML + DL + SHAP	Interpretable phishing detection

Chen et al. [9] presented a method based on data augmentation and a hybrid graph neural network to address the class imbalance problem in phishing detection in the Ethereum blockchain system. Wang et al. [1] proposed a novel method based on a dynamic co-associative semi-supervised graph neural network paradigm to address the problem of fraud detection in blockchain systems based on the evolution of graphs over time. Kasula et al. [7] presented a method based on graph neural networks for address classification in blockchain systems. Asiri and Somasundaram [6] studied graph convolution networks in the context of Bitcoin fraud detection and proved the portability of the GNN-based approach in various blockchain systems. Pocher et al. [5] developed a machine learning and graph-based forensics approach for anti-money laundering scenarios in Bitcoin transactions. Kılıç et al. [10] implemented classical machine learning for detecting blacklisted addresses in the Ethereum blockchain. Mounnan et al. [2] carried out an in-depth survey on deep anomaly detection methods in the context of blockchain systems. Shevchuk et al. [3] carried out an in-depth survey on the trends of anomaly detection in the context of blockchain systems and the limitation of the existing methods.

Despite these advancements, most of the existing studies are focused either on classical ML or DL individually. Only a few studies have attempted to incorporate explainability as an integral part of the detection process [2, 3]. In addition, comprehensive frameworks that address the detection of phishing attacks in the Ethereum blockchain environment, considering both balanced and imbalanced datasets and integrating the efficacy of ML, DL, and SHAP-based explainability methods [3, 9], are scarce. This study attempts to bridge the gap in the existing literature by presenting a comprehensive and interpretable framework.

3. Problem Formulation

The problem of phishing detection in Ethereum can be defined as a supervised binary classification problem for data in the blockchain [4, 8]. Let's denote the dataset of transactions as:

$$\mathcal{X} = \{x_1, x_2, \dots, x_n\}$$

Each transaction x_i is an instance of a confirmed transaction in the blockchain of the Ethereum network. Each transaction x_i can be defined by the following feature vector:

$$x_i = [v_i,\ b_i,\ t_i,\ f_i]$$

where v_i is the value of the transaction, b_i is the block height that represents the position of the transaction in the blockchain, t_i is the time associated with the transaction, and f_i represents engineered temporal features that include hour of day, day of month, month of year, etc. [5, 10]. These features include both transactional features that may differentiate legitimate and phishing-related transactions [4, 9].

Each transaction is associated with a binary class label, which can be defined as follows:

$$y_i \in \{0,\ 1\}$$

where $y_i = 0$ represents legitimate transactions, whereas $y_i = 1$ represents phishing-related transactions, as proposed in [4, 8]. The goal is to learn a function that can predict the class labels of the transactions as follows:

$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

The function should be able to minimize the misclassification error, as well as be robust to class imbalance, as phishing-related transactions are generally in the minority in real-world scenarios, as proposed in [9, 10].

One of the other requirements of this formulation, in addition to the prediction error, is interpretability, as proposed in [3, 5]. This is because, in the context of blockchain, the security of the system is of utmost importance, and hence, the model should be able to explain the decisions it is making, such that stakeholders can comprehend which features of the data are most important for phishing detection. [3, 5].

4. Proposed Framework

The proposed system has been designed to follow a well-structured pipeline to perform the task of phishing transaction detection on the Ethereum blockchain network.

The first step in the pipeline is the preprocessing of the transaction data. Here, the irrelevant attributes are removed from the transaction data. For example, the transaction hashes and wallet addresses are removed, as they are not relevant to the classification process. Missing data and duplicate entries are handled properly. The timestamp fields are converted to datetime format.

Next, feature engineering is carried out. Apart from basic features like transaction amount and block height, other time-based features like hour, day, and month are also created. This is because phishing behavior is usually not random but follows unusual time intervals.

The next step is training both machine learning and deep learning models. Conventional machine learning models like

Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine, K-Nearest Neighbors, etc., are used. Apart from that, a feedforward neural network is also used.

The performance of the model is checked using metrics such as precision, recall, F1 score, and ROC-AUC score. The experiments are performed on both balanced and imbalanced datasets.

Finally, SHAP is used for explainability purposes. Explainability is used to check which features affect the model, making it a reliable system.

5. Dataset Description

The dataset used in this study is comprised of actual Ethereum transactions that are categorized as legitimate or phishing-related. The data points in this dataset represent confirmed transactions.

Key attributes in this dataset include transaction value, block height, and timestamp. These attributes reflect both monetary and temporal aspects of users' behavior.

To assess the performance of the model properly, two datasets are considered. The first is a balanced dataset, in which both classes are equally represented. The second is an imbalanced dataset, which reflects real-world circumstances in which phishing-related transactions are infrequent.

Before processing and analyzing the dataset, duplicate values are removed and missing values are handled. The timestamp attribute is also structured for feature extraction.

6. Data Preprocessing

Data preprocessing is carried out to ensure that the data is clean and in an appropriate state for training. The data is cleaned by removing unnecessary fields like transaction ID and wallet address, as this information is not helpful for training.

Duplicates in the data are removed to ensure that there is no biased learning. The data is handled carefully to ensure that there are no missing values.

The timestamp is also converted from UNIX time to datetime format. This is helpful in deriving useful information from it.

Simple validation checks are also carried out to eliminate any outliers. The cleaned data is then ready for further processing.

7. Feature Engineering

Feature engineering is performed to enhance the quality of input data for these models. In addition to transaction value and block height, temporal features are also considered.

These are hour of the day, day of the month, and month of the year. These features can also be used to represent user behavior, as phishing activities are usually performed in an irregular manner.

Table 2
Final Feature Set Used for Classification

Feature	Description
BlockHeight	Position of transaction in blockchain
Value	Transaction amount
hour	Hour of transaction
day	Day of month
month	Month of year
weekday	Day of week

Transaction value is also considered an important feature, as phishing transactions may vary greatly compared to other normal transactions.

The final feature set includes both transaction and temporal information.

8. Machine Learning Models

To assess the performance of classical supervised learning approaches for Ethereum phishing detection, various machine learning algorithms are used [5, 6, 10]. These algorithms have been chosen as they implement different machine learning paradigms like tree learning, ensemble methods, margin methods, and distance methods. This allows for comprehensive evaluation of these algorithms and helps in choosing the best classifier for phishing detection problems [1].

A Decision Tree classifier is used as it is easy to understand and works well for problems with non-linear decision boundaries. A decision tree classifier divides the feature space into decision rules based on attribute values. Decision trees are computationally efficient and easy to understand. However, decision trees are prone to overfitting, especially in the presence of noisy data and feature interactions [3].

To overcome this drawback, Random Forest and Gradient Boosting ensemble techniques are used [1, 5]. Random Forest generates multiple decision trees using the bootstrap method with random selection of features, which reduces the variance of the model and improves the generalization performance [1]. Gradient Boosting grows multiple decision trees, with each tree attempting to correct the error of the previous tree in the ensemble [5]. This reduces the error of the ensemble by concentrating on misclassified samples, thus improving the performance of the model, which is suitable for structured tabular data such as blockchain transactions [1, 5].

The Support Vector Machine (SVM) classifier is also used to assess its potential to learn optimal hyperplanes in high-dimensional space [6]. SVM can learn optimal hyperplanes by maximizing the margin between classes, thus improving generalization performance, especially for well-separated classes. It can also learn non-linear relationships using kernel functions, although the performance of the classifier may be highly dependent on the selection of the kernel function [5].

Also, the K-Nearest Neighbors (KNN) algorithm is used, which is a distance classifier [10]. It labels the classes based on the majority class of the nearest data points in the feature space. KNN is computationally simple to implement, making no assumptions about the data distribution, but can be affected by scaling of the features and can be computationally costly for large data sets [3, 10].

All of these models offer a wide range of evaluations of conventional machine learning methods for phishing transactions detection, creating a strong baseline for the proposed use of the deep learning method [1, 5, 6].

9. Deep Learning Model

Besides the traditional machine learning models, a deep learning technique is also applied to identify complex non-linear relationships between the features of the transactions [1, 8, 9]. A feed-forward artificial neural network (ANN) architecture is developed as a multi-layer perceptron for binary classification of the Ethereum transactions.

The architecture of the proposed deep learning model is shown in Figure 1.

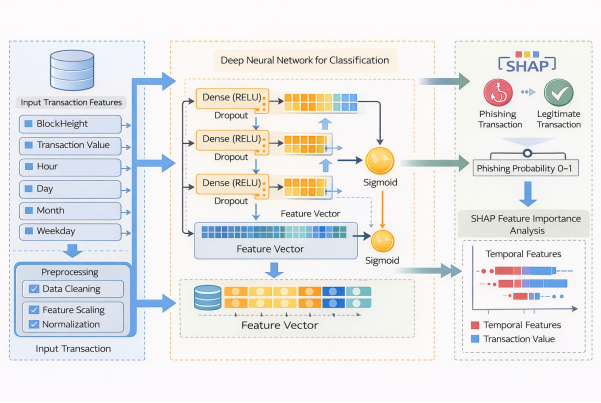

Figure 1: Deep Learning Architecture (Feedforward ANN)

Besides the traditional machine learning models, a deep learning technique is also applied to identify complex non-linear relationships between the features of the transactions [1, 8, 9]. A feed-forward artificial neural network (ANN) architecture is developed as a multi-layer perceptron for binary classification of the Ethereum transactions, as shown in Figure 1.

The input layer of the ANN accepts the engineered feature vector constructed from the transactional and temporal features. The features are then propagated through a number of fully connected hidden layers, where non-linear activation functions, such as ReLU (Rectified Linear Unit), are applied [1]. The hidden layers of the ANN are used to learn complex features that may not be learned by traditional machine learning models [9].

To further enhance generalization and reduce overfitting, various regularization techniques like dropout layers are added between dense layers [8]. The dropout technique randomly disables a number of neurons during training, which helps in learning more general feature patterns [1]. Normalization techniques like batch normalization can also be added to further stabilize training and speed up convergence [9].

Finally, an output layer is added with a single neuron and a sigmoid activation function that produces a probability score between 0 and 1 that reflects the probability that a transaction is associated with phishing activity [1]. The network is trained using binary cross-entropy loss and various optimization techniques that rely on gradients [8, 9].

This deep learning architecture improves the ability of the framework to capture complex patterns and dynamic phishing tactics, in addition to classical machine learning approaches [1, 3, 9].

Figure 2: Training vs. Validation Accuracy of Deep Learning Model

10. Experimental Setup

The experimental setup is proposed to test the performance of the proposed phishing detection framework in a real environment [1, 9]. The dataset is divided into two subsets, namely, the training dataset and the testing dataset, using an 80:20 split ratio. The training dataset is used for model learning, whereas the testing dataset is used for unbiased testing of the model performance on unseen data [5]. Both balanced and unbalanced dataset configurations are used to test the robustness of the proposed model [9, 10].

The experiments are conducted using Python-based machine learning libraries. Classical machine learning models are developed using popular frameworks to efficiently train the models for testing purposes [5, 6]. Similarly, the proposed deep learning model is developed using a neural network library, which supports dense models with gradient optimization techniques [1, 8].

The hyperparameters of the models are chosen using standard tuning techniques for fair comparison of the proposed models with existing models in the literature [1, 9]. Performance metrics are computed on the test dataset to test the effectiveness of the proposed models for classification purposes. During deep learning model training, regularization techniques are used to avoid overfitting of the model [8].

The experiments are carried out in a general computing environment with adequate processing power to process the data generated in the context of transactions [3]. This facilitates the reproduction of the experiments and proves

the possibility of implementing the proposed framework in the context of real-world blockchain security monitoring systems [1, 3].

11. Evaluation Metrics

To measure the effectiveness of the proposed phishing detection framework, various performance metrics are used [1, 5, 9]. These metrics offer a comprehensive analysis of classification behavior, especially in scenarios with class imbalance, where phishing transactions are in the minority.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy is used to measure the total number of accurate classifications. However, in scenarios with class imbalance, accuracy is not an appropriate measure, as high accuracy can be achieved by making incorrect classifications by always choosing the majority class [9, 10].

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision is used to measure the ratio of phishing-related transactions to the total number of phishing-related transactions predicted by the classifier. High precision means that there are fewer false positives, which is important to avoid false alarms [3].

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall, also known as sensitivity, measures the model's ability to correctly identify phishing transactions [1, 9]. High recall is essential in security applications, as missing a phishing transaction (false negative) can lead to financial loss [4].

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-score provides a harmonic mean of precision and recall, offering a balanced measure of detection performance when both false positives and false negatives are important [5, 9].

Moreover, the Receiver Operating Characteristic (ROC) Curve and the Area Under the Curve (ROC-AUC) are also used to assess the trade-off between true and false positive rates for different classification thresholds [1, 6]. The ROC-AUC is also used to compare different models regardless of the classification threshold [5].

12. Results and Performance Analysis

The experimental results prove that the ensemble learning methods, as well as the deep learning model, perform better in comparison to individual classical models [1, 9].

Table 3
Performance Comparison of Machine Learning Models

Model	Accuracy
Random Forest	0.9409
Decision Tree	0.9326
KNN	0.9187
SVM	0.8838
Gradient Boosting	0.8606
MLP	0.8465

Table 4
Classification Report for Best Model (Random Forest)

Class	Precision	Recall	F1-score	Support
Legitimate (0)	0.95	0.99	0.97	23,589
Phishing (1)	0.64	0.31	0.42	1,635
Accuracy		0.94		

This can be observed in particular in scenarios with class imbalance, where phishing transactions belong to the minority class [9, 10].

The tree ensemble models, such as Random Forest and Gradient Boosting, demonstrate high precision and recall, which means that they are able to detect phishing transactions with high accuracy, reducing false alarms in the process [1, 5]. Their capacity to use multiple decision trees for classification enables them to learn non-linear relationships well, reducing the possibility of overfitting, which is useful in dealing with complex and diverse transactional data [1].

The performance of the deep learning model is improved in detecting phishing transactions by leveraging complex interactions between features, which is useful in dealing with complex scenarios of phishing attacks with their ever-changing strategies [1, 8]. It obtains competitive F1 scores and ROC-AUC scores, which means that the model generalizes well to both balanced and unbalanced datasets, showing that neural networks can learn complex behavioral patterns associated with phishing attacks [1, 3, 9].

On the other hand, other models such as K-Nearest Neighbors and Decision Trees alone have relatively lower performance, especially in an imbalanced scenario [10]. These models may find it difficult to generalize, especially in cases of scarcity of phishing cases in the dataset, resulting in higher false negative rates [3, 9].

The above models indicate the importance of using ensemble models and deep learning models in detecting blockchain phishing scenarios [1, 9]. These models are capable of maintaining higher recall rates without compromising precision. In security-critical applications, it is vital to detect malicious activities. [3, 5].

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789/789 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step
              precision    recall  f1-score   support

           0       0.95      0.99      0.97     23589
           1       0.64      0.31      0.42      1635

    accuracy                           0.94     25224
   macro avg       0.80      0.65      0.70     25224
weighted avg       0.93      0.94      0.93     25224

[[23300   289]
 [ 1124   511]]
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Figure 3: Confusion Matrix of Best Performing Model (Random Forest)

13. Explainability Analysis Using SHAP

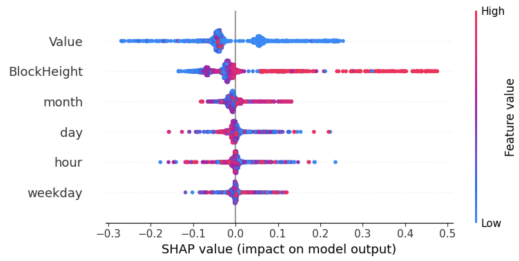

Figure 4: SHAP Feature Importance Summary Plot.

For the improvement of transparency and interpretability of the developed predictive models, the SHAP (SHapley Additive exPlanations) technique is utilized as a post-hoc explainability method [3, 5]. This is because the SHAP technique offers a theoretically grounded technique using cooperative game theory, where each feature is assigned an importance value that represents the contribution of each feature to the prediction [5].

The global explanation analysis is first performed to understand the feature influence [3, 5]. Figure 4 shows the SHAP summary plot, which represents the contribution of each feature to the prediction, indicating the most impactful features in detecting phishing attacks [5]. From the feature impact analysis, the transaction value and temporal features, such as hour and day, are identified to be impactful features in the classification process [5, 9], indicating that the attacks are likely to be unique in their transaction value and temporal patterns [4, 9].

Local explanations are also provided for individual predictions, which allow for an understanding of why certain transactions have been classified as phishing-related or legitimate [3, 5]. For example, high-value transactions with unusual timing may have a significant impact on the phishing classification [4, 5].

The integration of SHAP also resolves the interpretability issues associated with machine learning and deep learning techniques [2, 3]. Therefore, trust in automated decision-making is increased with the feature-level contributions provided by SHAP, which in turn enables the practical application of the detection framework in blockchain security monitoring [3, 5].

14. Discussion

To assess the contribution of individual components in the deep learning model, an ablation study is carried out. The results showed that removing dropout layers resulted in overfitting and degradation in validation accuracy [1, 8]. Removing the hidden layers resulted in degradation in detection accuracy, emphasizing the need for hierarchical feature learning [8, 9]. The use of a single dense layer resulted in limitations in modeling non-linear relationships [1]. This validates that the multilayer model with dropout improves performance and robustness.

14.1. Comparative Discussion

The comparative evaluation of classical machine learning models, deep learning models, and explainability techniques for phishing detection validates that each technique is effective in its own right in the proposed framework. [1, 3, 5].

Classical machine learning models show computational efficiency and ease of implementation [5, 6, 10]. Models such as Decision Trees, Random Forests, and Gradient Boosting show strong baseline performance and are suitable for structured tabular data [1, 5]. Ensemble methods show strong performance in terms of accuracy and have relatively low training complexity [1]. These models are also interpretable, especially in the case of decision trees and random forests, allowing partial understanding of the decision boundaries, making them suitable for large-scale implementations [3, 10].

Deep learning models show improved generalization and the capability of learning complex, nonlinear relationships between the features [1, 8, 9]. The feedforward neural network is suitable for modeling relationships between transactional and temporal attributes, which might not be easily captured by classical models [8]. This results in improved performance, especially in scenarios involving evolving phishing scenarios and unbalanced data sets [9]. However, these models also show relatively higher computational complexity and parameter tuning requirements [1, 3].

The inclusion of the integration of the explainability of the SHAP addresses the main challenge of most predictive models, which is the absence of transparency [2, 3, 5]. This is because the feature contribution quantification bridges the gap between the performance and transparency of the models [5]. This is crucial in the explanation of the decision-making process in security-related applications [3, 5].

The integration of ML, DL, and XAI is therefore effective in providing a balanced solution that ensures high accuracy, generalization, and trust in decision-making, which is crucial in the deployment of the blockchain-based phishing detection system [1, 3, 5].

14.2. State-of-the-Art Comparison

Existing phishing detection techniques that utilize graph neural networks and complex network models are reported in recent research studies [1, 4, 7, 8, 9]. Though these methods yield high accuracy in detecting phishing attacks, they are computationally expensive and lack explainability [2, 3].

The proposed ML-DL-XAI framework provides competitive accuracy with the features of transactions and also provides explainability [3, 5]. The proposed approach differs from the existing approaches [1, 4, 8, 9].

15. Limitations

Although the proposed framework shows promise in terms of phishing transactions, there are still some limitations to be considered. First, the study mainly considers different types of attributes related to transactions, such as the value of the transactions, block height, and time attributes [5, 10]. However, it is not fully representative of the complex relationships present in the blockchain ecosystem [4, 7, 8].

Another limitation of the proposed framework is that it is not considering graph-based attributes related to different types of wallet addresses [1, 4, 8]. Phishing activities often show different types of relationships between different addresses, fund transfer activities through different intermediary addresses, and different types of transactions [9].

Also, code analysis of the smart contracts is not taken into consideration [9]. The malicious smart contracts might have complex logic that tries to trick the users. The code analysis of the smart contracts might improve the detection capabilities of the framework [7]. The lack of semantic features at the contract level might limit the detection of phishing schemes that depend on the behavior of the smart contracts rather than the transactions themselves [9].

Another limitation of the framework is the dependence on the dataset. The framework is tested on the data that was previously labeled. The data might not be representative of the new phishing schemes and the evolving attack strategies that might be used by the attackers [3, 9]. The framework might require updates and learning mechanisms to be deployed in real-time scenarios [1, 3].

Lastly, although SHAP is interpretable, explanations might also incur additional computational costs, especially when dealing with large data sets or complex models [5]. The trade-off between interpretability and computational costs is an important consideration in large-scale blockchain monitoring systems [3].

16. Future Work

There are a number of ways that this framework can be extended to further improve phishing detection on the Ethereum network. One of these ways is the use of graph-based learning methods, as proposed in [1, 4, 7, 8]. The nature of blockchain-based transactions is that they are a network of interacting addresses, and this makes graph neural networks a useful tool for identifying relationships between them and detecting coordinated attacks.

Another area that this framework can be extended to include is the use of real-time blockchain monitoring, as proposed in [1, 3]. Although this paper focuses on the analysis of historical data, in a live system, it would be necessary to have a system for detecting suspicious activity as it occurs. This could potentially be achieved by incorporating online learning methods. [1, 3].

Future work can also include smart contract analysis, which can be accomplished through bytecode analysis, function-level analysis, and detection of malicious smart contract logic [7, 9]. An integrated approach using transactional, graph-level, and contract-level features can be used to improve the overall detection framework [9].

Adaptive learning can be integrated to deal with the ever-changing phishing schemes [3]. Incremental updating of models, as well as anomaly detection, can be used to improve the overall detection performance over time [1, 3]. Moreover, optimization of explainability techniques to minimize computational costs while maintaining interpretability can improve the scalability of the framework [5].

17. Conclusion

This paper proposes an all-encompassing framework for phishing detection in Ethereum using classical machine learning, deep learning, and explainable artificial intelligence techniques, as mentioned in. By using features obtained from real-world blockchain data, the proposed system is capable of effectively distinguishing phishing transactions from normal transactions. The comparative analysis of the proposed models shows that ensemble learning techniques and neural network-based models have high predictive accuracy in class imbalance conditions.

Moreover, in addition to high accuracy, the proposed system also considers the importance of model explainability using SHAP-based explainability techniques. Using explainability techniques, it is evident that the value of transactions is an essential feature in phishing detection, which is a significant insight into model behavior. This is an essential feature in model development, as transparency is a significant factor in model development in blockchain-based environments.

The importance of using predictive models with explainability is also emphasized in the study, which can solve both performance- and usability-related issues. Classical machine learning models provide efficiency and robustness to the models, whereas deep learning models improve the generalization of the models, and explainable AI connects the dots between the complexity of the models and the level of trust placed in them by the users. Thus, all the components of the proposed framework provide a well-balanced solution for blockchain security monitoring.

The proposed framework can contribute to the development of an intelligent, transparent, and scalable phishing detection system, which can be achieved by incorporating ML, DL, and XAI models to improve the security of the decentralized financial system from phishing attacks.

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References

[1] Y. Wang, Q. Zheng, X. Li, L. Wang, and L. Lin, "CoSemiGNN: Blockchain fraud detection with dynamic graph neural networks based on co-association of semi-supervised," *Expert Systems with Applications*, vol. 298, 2026.

[2] O. Mounnan, O. Manad, L. Boubchir, A. El Mouatasim, and B. Daachi, "A review on deep anomaly detection in blockchain," *Blockchain: Research and Applications*, vol. 5, no. 2, 2024.

[3] R. Shevchuk, V. Martsenyuk, B. Adamyk, V. Benson, and A. Melnyk, "Anomaly detection in blockchain: A systematic review of trends, challenges, and future directions," *Applied Sciences*, vol. 15, 2025.

[4] Z. Zhang, T. He, K. Chen, B. Zhang, Q. Wang, and L. Yuan, "Phishing node detection in Ethereum transaction network using graph convolutional networks," *Applied Sciences*, vol. 13, 2023.

[5] N. Pocher, M. Zichichi, F. Merizzi, M. Z. Shafiq, and S. Ferretti, "Detecting anomalous cryptocurrency transactions: An AML/CFT application of machine learning-based forensics," *Electronic Markets*, vol. 33, 2023.

[6] A. Asiri and K. Somasundaram, "Graph convolution network for fraud detection in bitcoin transactions," *Scientific Reports*, vol. 15, 2025.

[7] V. K. Kasula, A. R. Yadulla, M. Yenugula, B. Konda, and S. Ayyamgari, "Blockchain address classification using graph neural networks for enhanced security and analytics," *Discover Artificial Intelligence*, vol. 5, 2025.

[8] J. Kim, S. Lee, Y. Kim, S. Ahn, and S. Cho, "Graph learning-based blockchain phishing account detection with a heterogeneous transaction graph," *Sensors*, vol. 23, 2023.

[9] Z. Chen, S.-Z. Liu, J. Huang, Y.-H. Xiu, H. Zhang, and H.-X. Long, "Ethereum phishing scam detection based on data augmentation method and hybrid graph neural network model," *Sensors*, vol. 24, 2024.

[10] B. Kılıç, A. Şen, and C. Özturan, "Fraud detection in blockchains using machine learning," in *Proc. IEEE Conf.*, 2021.

[11] M. Alharbi, A. Alzahrani, and F. Alotaibi, "Blockchain fraud detection using explainable machine learning techniques," *IEEE Access*, vol. 11, pp. 125678–125692, 2023.

[12] H. Zhou, Y. Wang, and J. Liu, "Deep learning-based fraud detection in Ethereum blockchain transactions," *Future Generation Computer Systems*, vol. 144, pp. 12–25, 2023.

[13] S. Rana, P. Gupta, and A. K. Singh, "Explainable AI for blockchain fraud detection: A SHAP-based approach," *Expert Systems with Applications*, vol. 213, 2023.

[14] J. Park, S. Kim, and H. Lee, "Temporal graph neural networks for cryptocurrency fraud detection," *Information Sciences*, vol. 646, pp. 119–135, 2024.

[15] L. Zhang, Y. Xu, and T. Li, "Hybrid machine learning framework for detecting phishing attacks in blockchain systems," *Computers & Security*, vol. 134, 2024.

[16] R. Kumar and S. Tripathi, "A survey on blockchain security and fraud detection using AI techniques," *ACM Computing Surveys*, vol. 56, no. 4, 2024.

[17] Y. Li, Q. Chen, and X. Zhao, "Anomaly detection in blockchain using deep autoencoders," *IEEE Transactions on Network Science and Engineering*, vol. 11, no. 2, pp. 1450–1462, 2024.

[18] D. Singh and V. Kumar, "Machine learning-based phishing detection in decentralized finance platforms," *Journal of Information Security and Applications*, vol. 75, 2023.

[19] K. Mehta, S. Sharma, and R. Gupta, "Detecting fraudulent transactions in blockchain using ensemble learning," *Applied Soft Computing*, vol. 138, 2023.

[20] T. Nguyen, M. Pham, and L. Vo, "Explainable deep learning for financial fraud detection in blockchain systems," *Knowledge-Based Systems*, vol. 278, 2024.